National University of Computer & Emerging Sciences, Karachi

Spring 2026 CS-Department

Software Engineering

Project Description

Group Length : [min 2 and max 3]

1. **The objective of this project is to apply Software Engineering principles to design, analyze, and implement a system focusing on:**
    a. Choose one clear technical focus area, such as:
        i. Load Balancer Design
            1. Implement and compare Round Robin vs Least Connections
            2. Dynamic load balancing simulation
            3. AI-based traffic prediction for smart routing
        ii. Server Architecture Optimization
            1. Monolithic vs Microservices comparison
            2. Scalable backend architecture design
            3. Caching strategies (Redis-based simulation)
            4. API Gateway architecture
        iii. Backend Performance Optimization
            1. Database indexing impact analysis
            2. Query optimization benchmarking
            3. Concurrency handling
            4. Multithreading vs asynchronous processing
        iv. Design Pattern Implementation
            1. Apply Singleton, Factory, Observer, Strategy, MVC, etc.
            2. Show how pattern improves maintainability or scalability
            3. Refactor a poorly designed system using proper patterns
        v. AI-Based Optimization
            1. AI-based load prediction
            2. Resource allocation prediction
            3. Smart scheduling algorithm
            4. Performance anomaly detection
2. **Deliverables:**
    a. Working Project
        i. Functional prototype (Web-based or AI-based)
        ii. Code repository
        iii. Clear demonstration of optimization/design technique
    b. SRS Report (ieee-formate)
3. **Rubrics:**

Problem Definition	Technical Depth	System UML	Testing	Performance Comparison/validation	SRS	USE of Design & Architecture pattern	Total
5	10	5	5	5	5	5	40

4. Timeline

Week 7: Proposal submission (1-page idea approval)

DEADLINE: Week 15: Final submission (Report + Demo + Code)

Only one submission required from group leader